

07 - MeF Transmission Module

Overview

The MeF (Modernized e-File) module is the primary engine for federal tax return transmission. It handles schema validation, digital signature, package assembly, submission, and acknowledgment processing.

MeF System Components

Component	Function
Schema Validator	Validates XML return against IRS MeF schema
Return Packager	Assembles submission ZIP (return XML + attachments)
Digital Signer	Applies ERO PIN and EFIN signature
Submission Router	Routes to A2A engine
Acknowledgment Processor	Parses IRS ACK and updates status
Error Parser	Maps IRS error codes to fix instructions

MeF Schema Versions

Tax Year	Schema Version	Status
2025	MeF 2025v1.0	Current
2024	MeF 2024v5.0	Supported
2023	MeF 2023v4.0	Supported
2022	MeF 2022v3.0	Legacy Support
2021	MeF 2021v2.0	Expired

Return Package Structure

```
submission_{UUID}.zip
|- manifest.xml
|- return.xml
```

```
| - attachments/
|   | - w2_001.xml
|   | - 1099_001.xml
|   \- supporting_docs/
\ - signature.xml
```

Signature Block

```
<ero:Signature xmlns:ero="http://www.irs.gov/ero">
  <ero:EFIN>748335</ero:EFIN>
  <ero:ERO_PIN>[5-digit self-select PIN]</ero:ERO_PIN>
  <ero:SignatureDate>2026-04-15</ero:SignatureDate>
  <ero:IPAddress>[Preparer workstation IP]</ero:IPAddress>
  <ero:SoftwareID>[PractitionerHub MeF ID]</ero:SoftwareID>
</ero:Signature>
```

Business Rules Validation

Rule Category	Examples
Filing Status	MFS not allowed if state requires MFJ
Dependent Validation	SSN must not be duplicate across returns
Income Thresholds	EIC phase-out limits enforced
Form Attachment	Schedule required when line exceeds threshold
Date Validation	Tax year must match period begin/end dates
Payment Routing	Routing number must pass check-digit validation

MeF API Endpoints (Production)

Operation	Endpoint
Submit Return	POST https://la.www4.irs.gov/a2a/
Get Acknowledgment	GET https://la.www4.irs.gov/a2a/ack/{submissionId}
Validate Schema	Internal offline schema check

Performance Benchmarks

Metric	Target	Current
Schema validation time	< 500ms	~320ms
Package assembly time	< 2s	~1.1s
Submission to ACK (Accepted)	< 24hr	~4-6hr avg
Submission to ACK (Rejected)	< 2hr	~45min avg

Metric	Target	Current
Batch throughput	100 returns/batch	Meets target